INCH-POUND

MIL-S-14799B (AR)
AMENDMENT 5
2 March 1989
SUPERSEDING
AMENDMENT 4
12 November 1988

MILITARY SPECIFICATION

SAFETY AND ARMING DEVICE, GUIDED MISSILE: M114
LOADING, ASSEMBLING AND PACKING

This Amendment forms a part of Military Specification MIL-S-14799B (AR), dated 4 September 1973, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

3.4.1, Delete paragraph in its entirety and substitute the following:

"3.4.1 Armed Condition. The electrical resistance between the brown lead of the two lead wires and the red lead of the three lead wires shall not be less than 750 ohms nor more than 15000 ohms when tested at ambient temperature."

PAGE 3

* 3.5.1, Delete "Not less than 325 nor more than 429 milliseconds." and replace with "Not less than 320 nor more than 429 milliseconds."

* 3.5.2, Delete "Not less than 325 nor more than 483 milliseconds." and replace with "Not less than 320 nor more than 483 milliseconds."

AMSC N/A FSC 1336

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PAGE 4

3.8 6, Add new paragraph as follows:

"3.8.6 Shock. The assembly with shorting screw inserted shall withstand a shock of 350 ± 50 g with a pulse duration of 700-800 microseconds as specified in 4.4.13."

* 3.13, Add new paragraph as follows:

"3.13 Leaf spring contact. The leaf spring contact tip in the "safe" condition shall be inspected in accordance with 4.3.3.2.14."

PAGE 5

4.2.1, Add reference to new drawing "x. 12579605."

PAGE 7

4.3.1, Add at end of paragraph the following:

"Two lots of detonators from one manufacturer may be mixed in an S&A lot when one detonator lot is completely depleted and a second detonator lot is required to complete the S&A Device lot."

4.3.1.1 thru 4.3.1.4, Add new paragraphs as follows:

"4.3.1.1 Coupons for polymer coating. In order to perform adhesion testing and dimensional analysis, a test coupon shall be supplied with each lot of parts to be processed with Coating, Polymer Film Lubricant. A lot of parts is defined as parts of the same base metal to be coated under the same conditions and by means of the same coating process (Dwg. No. 9231467, Note 6; Dwg. No. 9231470, Note 5). For requirements reference drawing notes to see coating, polymer film lubricant drawing.

4.3.1.1.1 Types of coupon. The dimensions by type shall not deviate from any dimension by more than 25%.

4.3.1.1.1.1 Rectangular. The dimensions shall be 1.5" x 0.5"x 0.04.

4.3.1.2 Codes and material.

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4.3.1.2.1 Codes. Each test coupon shall be marked with two identifying codes.

Table IV. Material Codes

Code	Material Description	Material Spec.
C1	Brass, strip, copper alloy 260, 268 half hard or full hard	ASTM B36

4.3.1.2.1.1 Code. The second identifying code is left to the discretion of the Contractor. It must be adequate to associate the test coupon to the represented lot of parts. The two identifying codes are to be stamped on the coupons away from the center of the coupon so as not to interfere with the surface of interest in the adhesion test. Deformation of the surface in the area of the stamping is permissable.

4.3.1.2.2 Material. Each coupon shall be of the identical material and in the same condition as the part(s) it represents. This is to include any treatments such as passivation or heat treatment to which the parts are subjected to before the application of the polymer film coating.

4.3.1.3 Thickness inspection.

4.3.1.3.1 Microscopic method. The coating thickness shall be determined in accordance with ASTM-B487. A minimum of five (5) measurements are to be taken and the average of these measurements will determine the coating thickness. Otherwise an instrument method may be employed.

4.3.1.3.2 Instrument method. The instrument, its calibration, and its operation shall be such that the coating thickness will be determined within ± 10 percent of its true thickness.

4.3.1.4 Adhesion test. Rectangular test coupons will be bent 180 ± 10 degrees (relaxation to a lesser angle after initially reaching 180 ± 10 degrees is permitted) about a mandrel with a diameter no greater than four (4) times the thickness of the test coupon. Cylindrical test coupons are to be bent 90 ± 10 degrees (relaxation to a lesser angle

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after initially reaching 90 ± 10 degrees is permitted) about a mandrel with a diameter no greater than four (4) times the diameter of the test coupon. The coupon is then to be examined at four (4) power magnification in the outer radius of the bend for flaking or separation of the coating from the base metal. The criteria for acceptance shall be that the coupon exhibits no evidence of flaking or separation of the coating from the coupon.

If the test coupon fractures during the bending process before the desired angle is reached, an alternate inspection is to be employed. First the area of the fracture is to be inspected for evidence of flaking or separation of the coating. In addition, a sharp instrument (Exacto Slimstyle Knife Catalog No. H3-3201, Blade No. H3-416, or approved equivalent) shall be used to scratch a cross hatched pattern on the larger of the two vertical lines and two horizontal lines extending the full length and width of the test coupon. The pattern is to consist of two vertical lines and two horizontal lines extending the full length and width of the coupon piece.

The scratched lines are to be sufficiently deep to reach the base metal. The lines should appear as continuous lines with no evidence of flaking or separation of the coating. Failure of the test coupon to meet either the thickness or adhesion criterion shall be cause for rejection of all parts which the test coupon represents."

PAGE 9

4.3.2.4, Delete Minor 201, and change Minor category AQL from "1.00" to "0.65."

4.3.2.5, Delete Major 101 and change AQL from "1.0" to "0.65" percent.

PAGE 10

4.3.2.7:

Add Major 104 as follows:

"104 Detonators leads not across head of pin latch (as viewed through pin, latch hole in rotor) ... Visual 07005.

Add Major 105 as follows:

105 Leaf spring contact ... Visual."

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PAGE 12

* 4.3.2.12, Add Major 102 as follows:

"102 Inspection of application of adhesive, dwg. (Note 3)."

PAGE 13

4.3.2.13, Delete Major 106.

PAGE 14

* 4.3.2.15, Add Major 103 as follows:

"103 Inspection of application of adhesive, dwg. (Note 2)."

PAGE 15

* 4.3.2.19, Add Major 105 as follows:

"105 Inspection of application of adhesive, dwg. (Note 5)."

* 4.3.2.20, Add Major 104 as follows:

"104 Leaf spring contact improperly positioned...100% ...3.13...Visual."

PAGE 16

* 4.3.2.22, Add Major 102 as follows:

"102 Inspection of application of adhesive, dwg. (Note3)."

PAGE 18

4.3.2.27 thru 4.3.2.28, Add new paragraphs as follows:

"4.3.2.27 Spacer, Pallet (see dwg. 9295018 covering a detail of 9231438).

Categories	Defects	Method of Insp.	Code No.
Major: 101	AQL 0.40 Percent Width of spacer, max.	Gage	32001

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4.3.2.28 Leaf "G" Sensing Assembly (see dwg. 9311016)

Categories	Defects	Method of Insp.	Code No.
Major: 101	"G" weight not flush or below leaf after riveting	Visual	34001"

4.3.3.1.2, Delete this paragraph in its entirety.

* 4.3.3.1.11, Add new paragraph as follows: "Visual inspection for proper application of adhesives (see 4.4.14).

PAGE 19

Table I, Test VI Change:

"VI Electrical resistance	4.4.6.1	Major	23017"
to			
"VI Electrical resistance	4.4.6.1	For Info. only	23017"

PAGE 21

4.3.3.2.1, line 2: Change "...spring, helical, torsion - rotor (dwg. (9231492)..." to "...spring, grounding, torsion - rotor (dwg. 9231492)..."

4.3.3.2.2, Delete this paragraph in its entirety.

PAGE 22

4.3.3.2.8, Delete this paragraph in its entirety and substitute the following:

"4.3.3.2.8 Environmental functioning and destructive testing (see 3.8). Testing shall be in accordance with Table I using equipment in accordance with 4.3.4. Sampling shall be in accordance with Figures 2 through 5. Temperature conditioning shall be at the required temperature plus or minus 5 degrees F. for two (2) hours, minimum.

The lot shall be rejected if:

a. Any unit fails the acceptance criteria specified in MIL-STD-331.

b. Two (2) or more units fail to arm.

c. Any unit fails the minimum arming time requirement.

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d. Any unit fails the electrical functioning requirement.

e. The sample average plus or minus 2 sigma fails the arming time requirements of 3.5.1 at ambient.

f. The sample average plus or minus 2 sigma fails the arming time requirements of 3.5.2 at low temperature (-25 degrees F)

In the event of one (1) electrical function failure at minus 25 degrees F. (in lots of 1201-10000 units) 60 additional units shall be selected, conditioned at minus 25 degrees F. for 2 hours minimum, and then functioned. If a total of two (2) electrical function failures occur at cold temperature, the lot shall be rejected. In the event of an electrical function failure at minus 25 degrees F. for lots less than 1201 units, the lot shall be rejected.

In the calculation for sample average and sample standard deviation, two long arming times may be excluded from the total lot test sample, provided neither of the long times exceed 500 milliseconds. The two times may be excluded from either the cold test group or the ambient test group, or one long time may be excluded from each test group."

PAGE 27

4.3.3.2.13, Add new testing paragraph as follows:

"4.3.3.2.13 Shock - major defect. This test shall be performed on samples specified in 4.4.13 and Figures 1 thru 5, as applicable, using equipment in accordance with 4.3.4. If any assembly fails the requirement the lot shall be rejected."

* 4.3.3.2.14, Add new paragraph as follows:

"4.3.3.2.14 Visual examination for leaf spring leaf tip position. (see figure 9 and 10) This examination shall be performed 100 percent prior to shipping."

4.4.2, Delete this paragraph in its entirety.

4.4.4.2, 1st sentence: Change "an electrical pulse of two (2) amperes maximum" to "an electrical pulse of 2.0 to 2.1 amperes."

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PAGE 28

4.4.5, Delete this paragraph in its entirety and substitute the following:

"4.4.5 Non-Arming. The Safety and Arming Mechanism Assembly shall be mounted to the test equipment in accordance with 4.3.4. Orientation of the assembly shall be such that acceleration simulates missile flight. The equipment shall be subject the assembly to a constant load of 10.0 g's + 0.0 g's - 1.0 g's at the center of gravity of the assembly for a minimum of three (3) seconds. Upon attaining the constant "g" load condition, the "g" sensing leaf shall be unlocked by mechanically removing the restraint from the "g" leaf. The "g" sensing leaf shall remain unlocked until the safe condition has been verified. The safe condition shall be verified by orientation of the rotor shaft and "g" sensing leaf retaining the latching leaf. This test shall be performed at ambient temperature."

PAGE 29

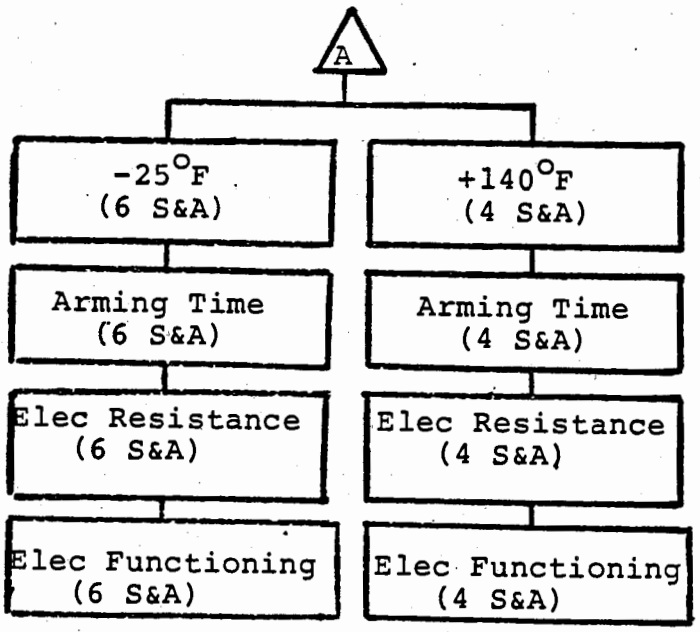
* 4.4.9, Delete in its entirety and substitute the following:

"4.4.9 Detonator electrical functioning. The detonator shall be temperature conditioned at -25°F + 0°F -10°F for one (1) hour minimum. Connect the detonator to the firing circuit specified. The energy from a. .56 microfarad capacitor ± 10% charge to 45 volts maximum shall be discharged to fire the detonator. Detonators that fail to fire on the first attempt shall be classed as failures. Functioning time may be recorded for information. If removal of detonators from the temperature chamber is required to accomodate functioning, the time from removal of the detonator from the temperature chamber until functioning shall not exceed ten (10) seconds. Units removed for longer than ten seconds must be returned to the temperature chamber for one hour minimum before testing."

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Figure 1 change,

From:



To:

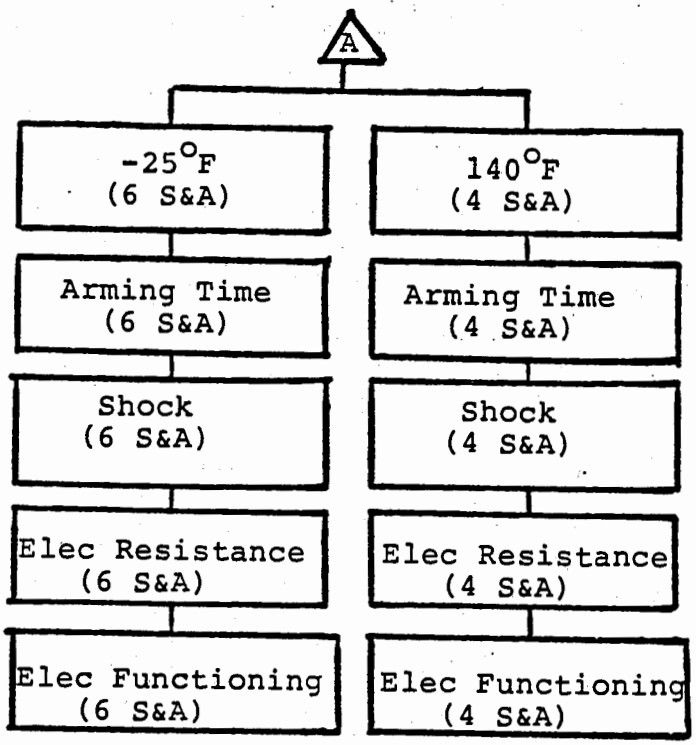
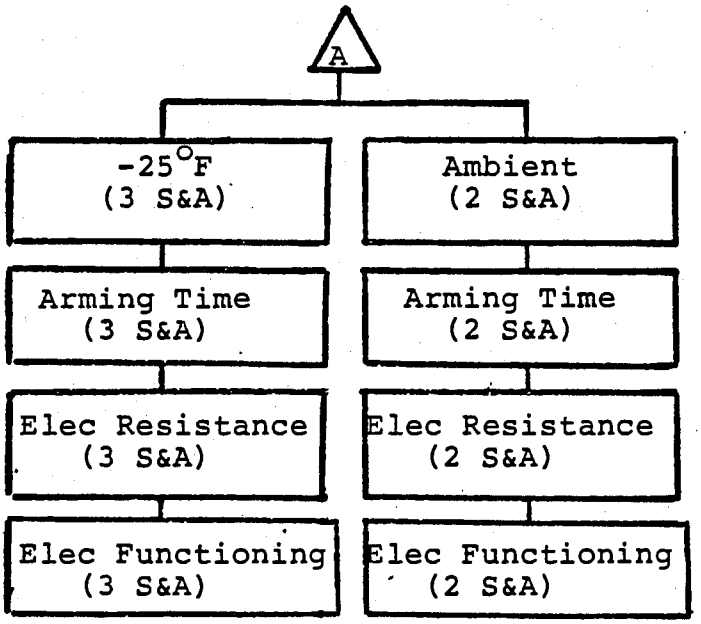


Figure 1

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Figure 2 change,

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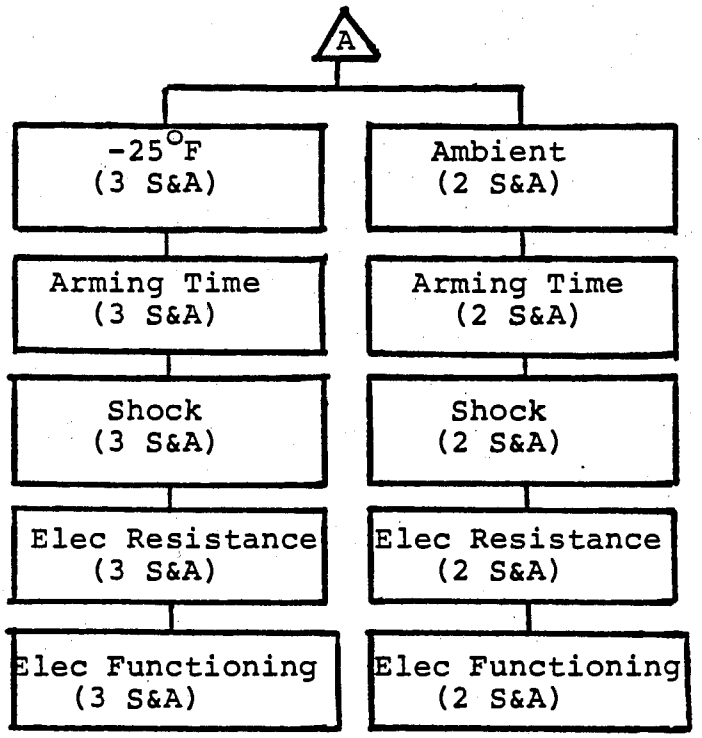
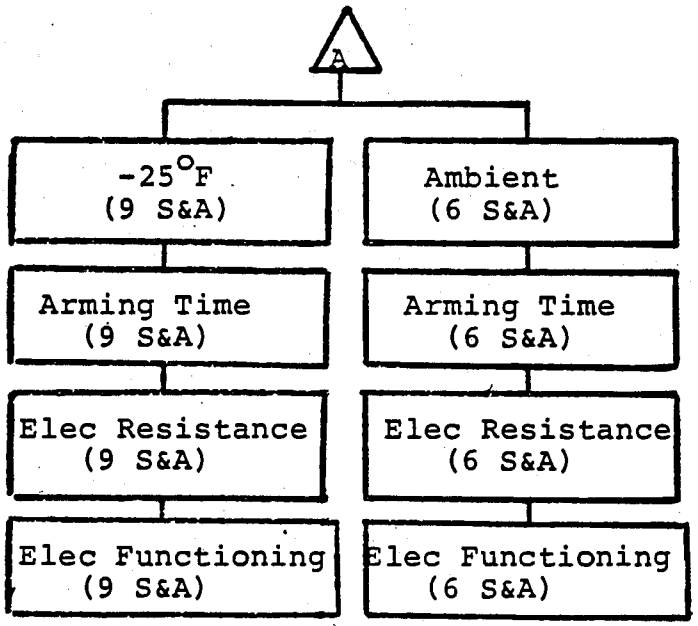


Figure 2

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Figure 3 change,

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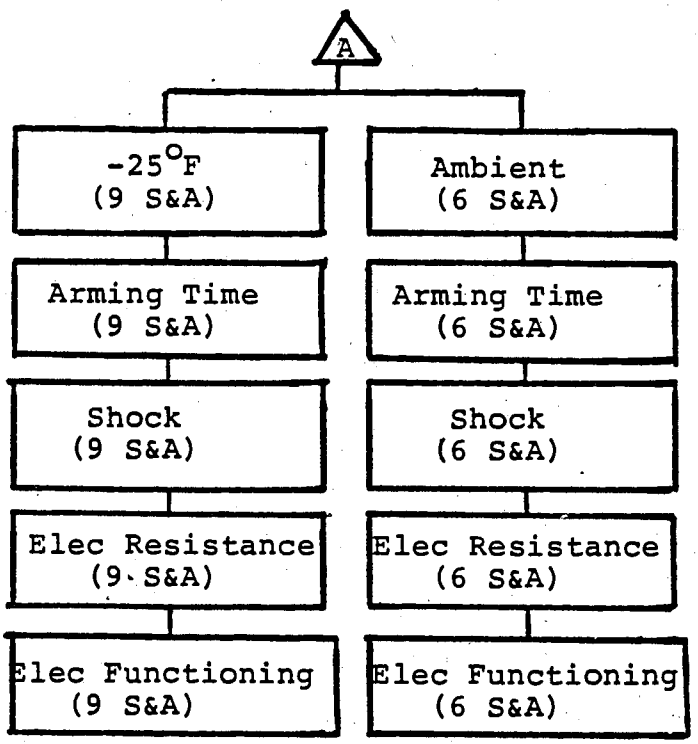
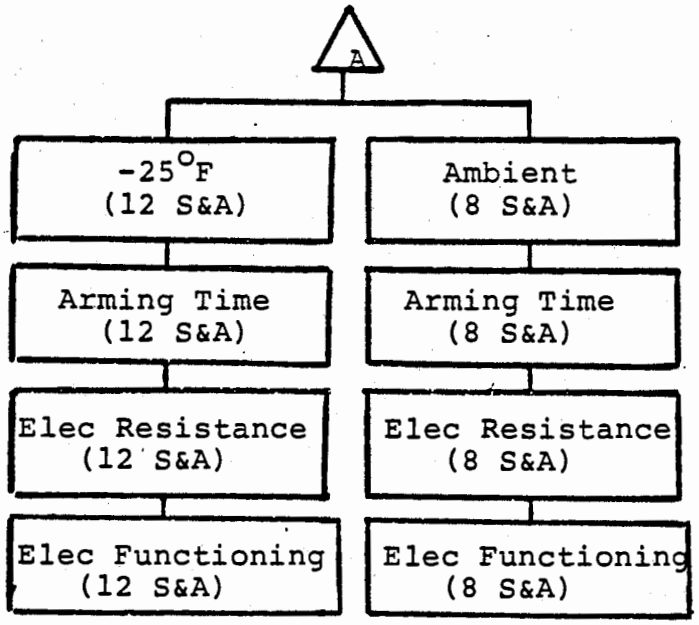


Figure 3

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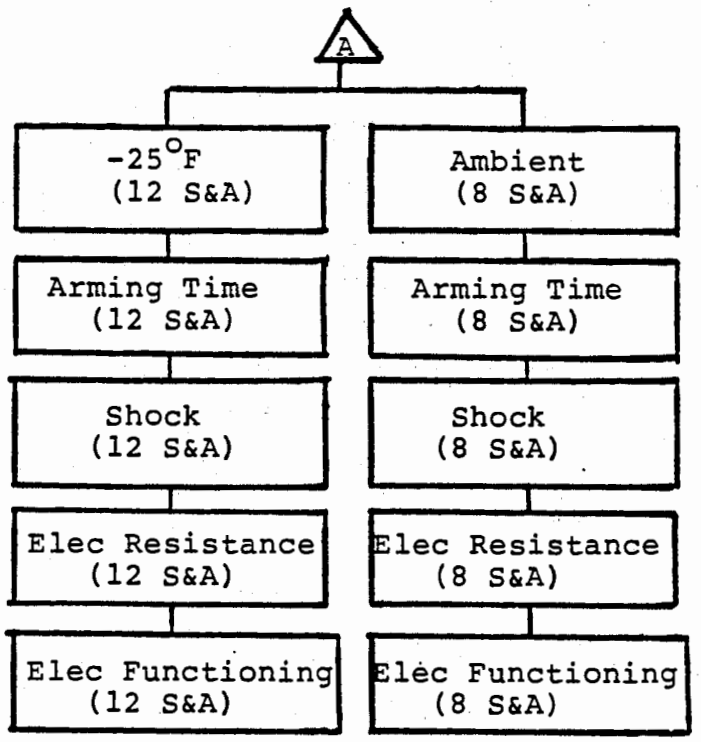
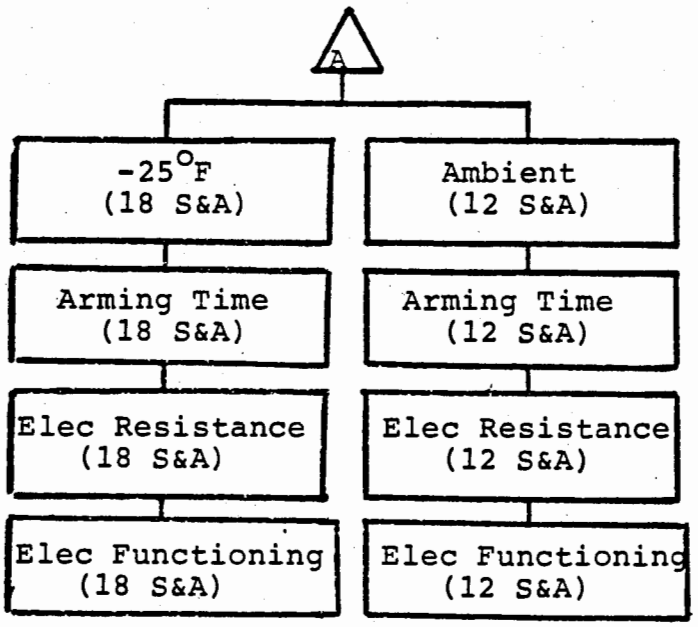


Figure 4

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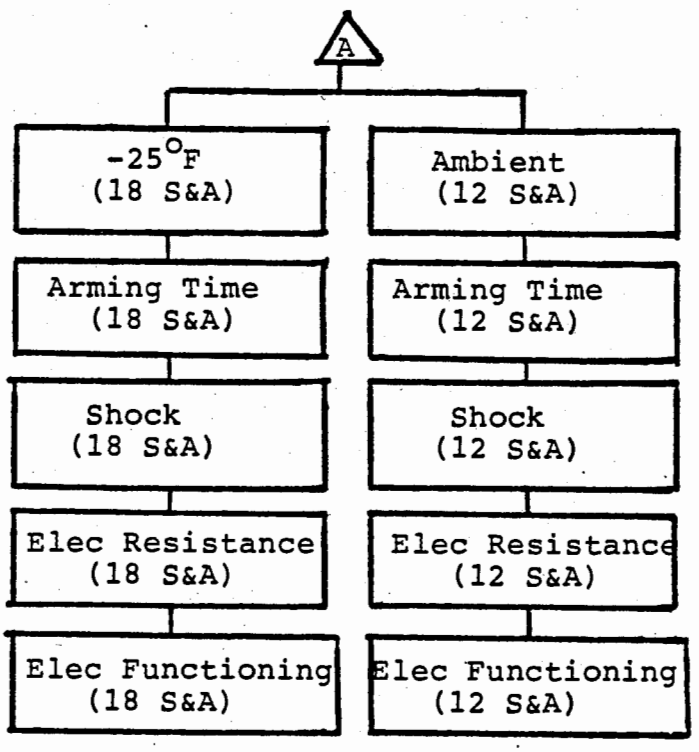
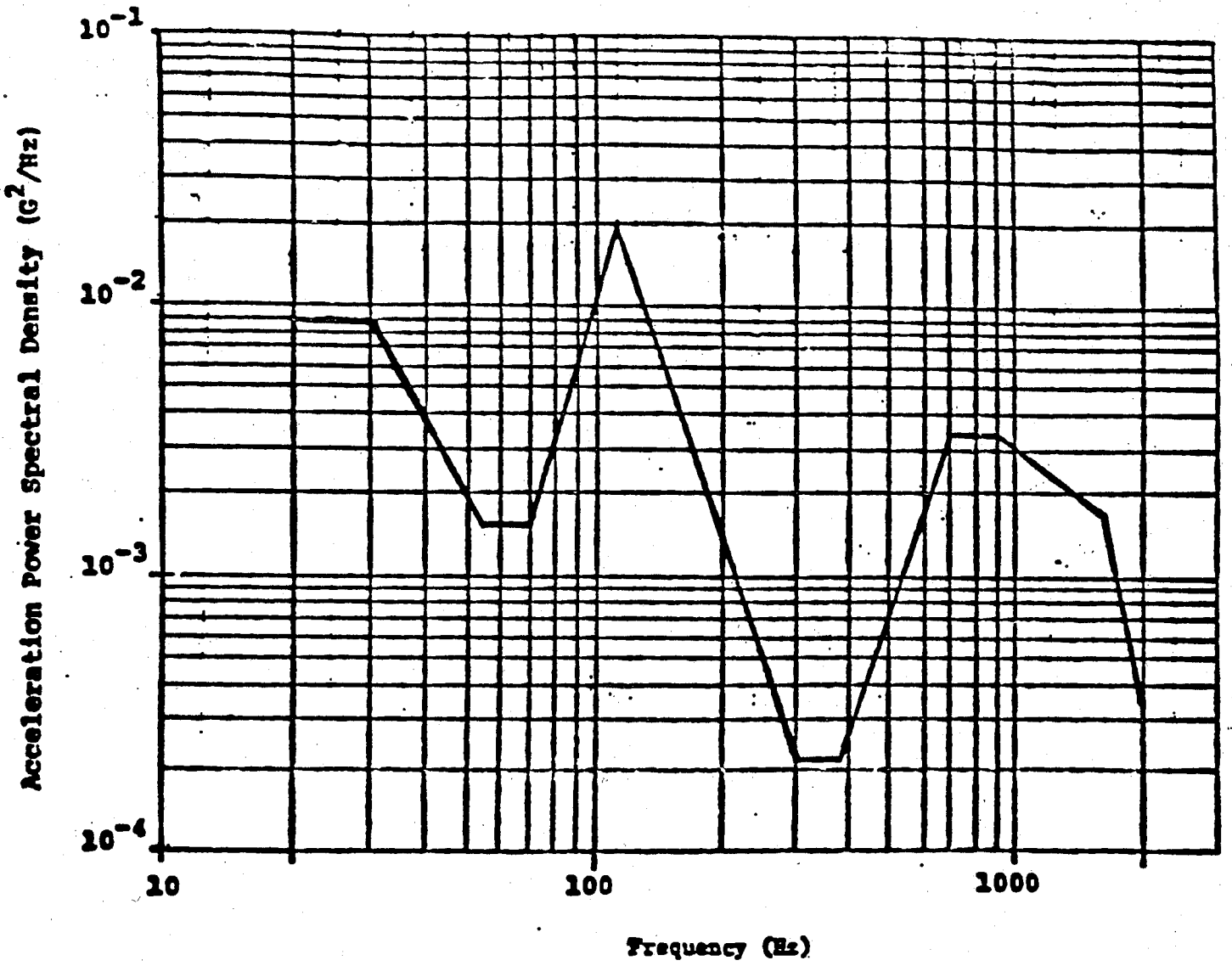


Figure 5

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* Figure 6, Delete in its entirety and substitute:



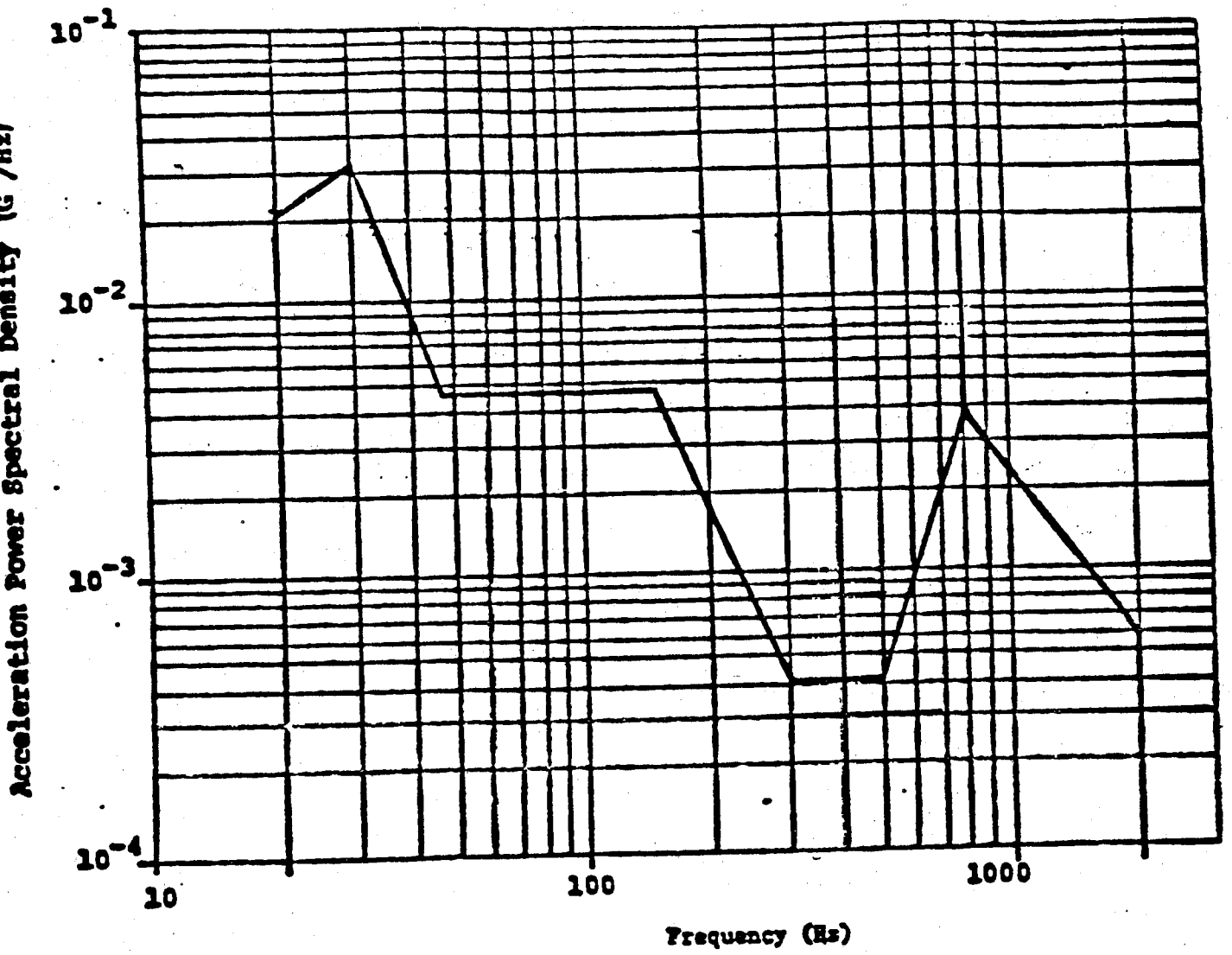
Level: 2.080 G RMS.

Transverse Axis Flight Vibration

Figure 6

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* Figure 7, Delete in its entirety and substitute:



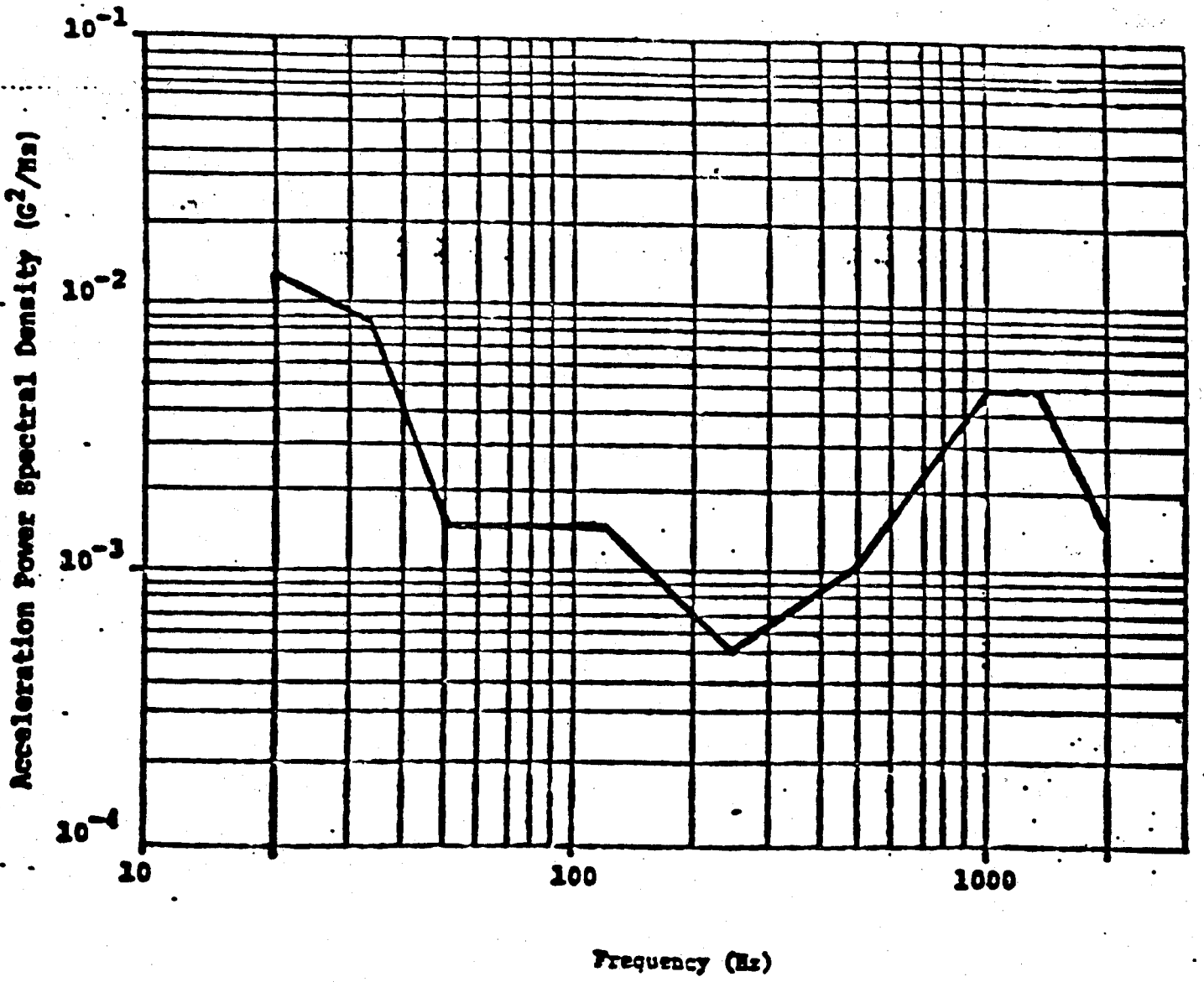
Level: 1.900 G RMS

Verticle Axis Flight Vibration

Figure 7

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* Figure 8, Delete in its entirety and substitute:



Level: 2.340 G RMS.

Longitudinal Axis Flight Vibration

Figure 8

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* Add figure 9

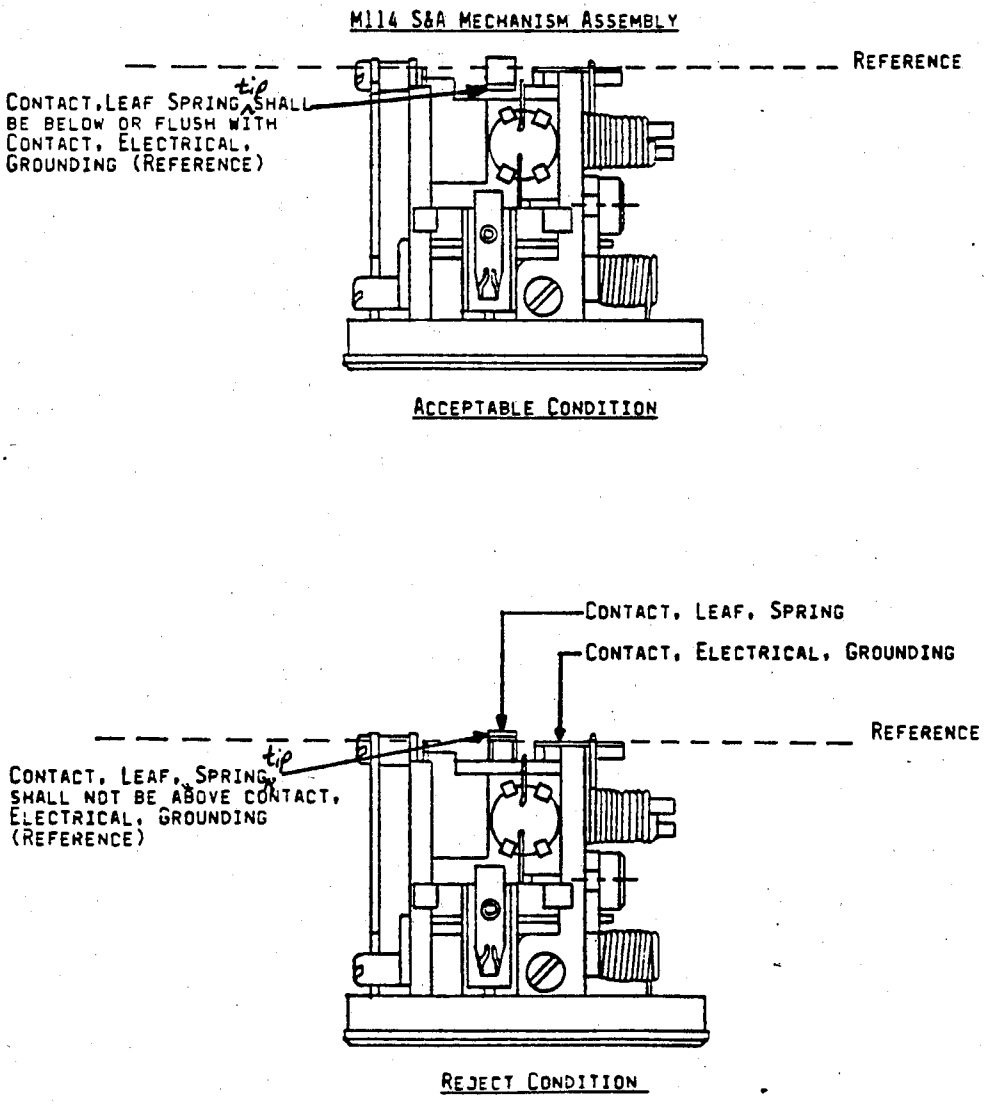


Figure 9. Acceptance criteria for leaf spring contact (front view)

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* Add figure 10

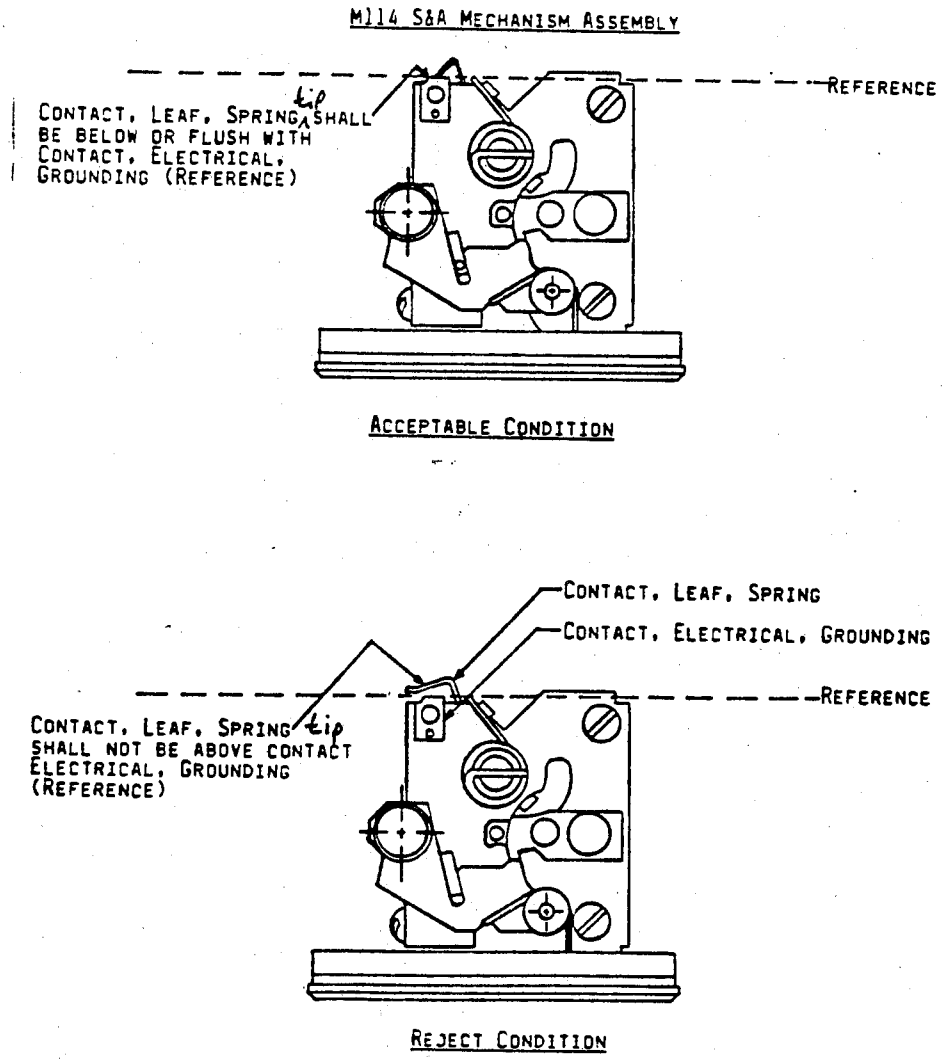


Figure 10. Acceptance criteria for leaf spring position (side views).

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4.4.13, Add new test procedure as follows:

"4.4.13 Shock. The safety and arming mechanism assembly shall be conditioned at the temperatures specified in Figures 1 thru 5 as applicable and then mounted to the test equipment in accordance with 4.3.4. Each assembly shall be subjected to the following orientations: longitudinal axis, leaf spring contact down and the escapement assembly down (90° from leaf spring down). The equipment shall subject the assemblies to a shock of 350 g ± 50 g with a pulse duration of 700-800 microseconds in 2 directions."

* 4.4.14, Add new paragraph as follows:

"4.4.14 Visual inspection for proper application of adhesive. The assemblies shall be inspected for proper application in compliance with the drawings. This inspection may require a destructive test."

PAGE 34

5.1.1, Change "level A" to "level C."

5.2.1, Change "level A" to "level C."

The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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